

THEORY



Training Program 01

Advanced NLP & Large Language Models (LLMs)



CONFIANZA23
Inteligencia y Seguridad

Training Program 01

M01: Text Preprocessing

M02: Word and Sentence Embeddings

M03: Transformer Architectures

M04: Entity-Centric NLP

M05: Applied NLP Tasks

M06: Multimodal NLP

M07: LLM Fundamentals

M08: Prompt Engineering

M09: Fine-Tuning & PEFT

M10: RAG & Real-World Projects



MODULE M01

TEXT PREPROCESSING

CHAPTER M04

Entity-Centric NLP



CONFIANZA23
Inteligencia y Seguridad

4.1 Introduction to Entity-Centric NLP

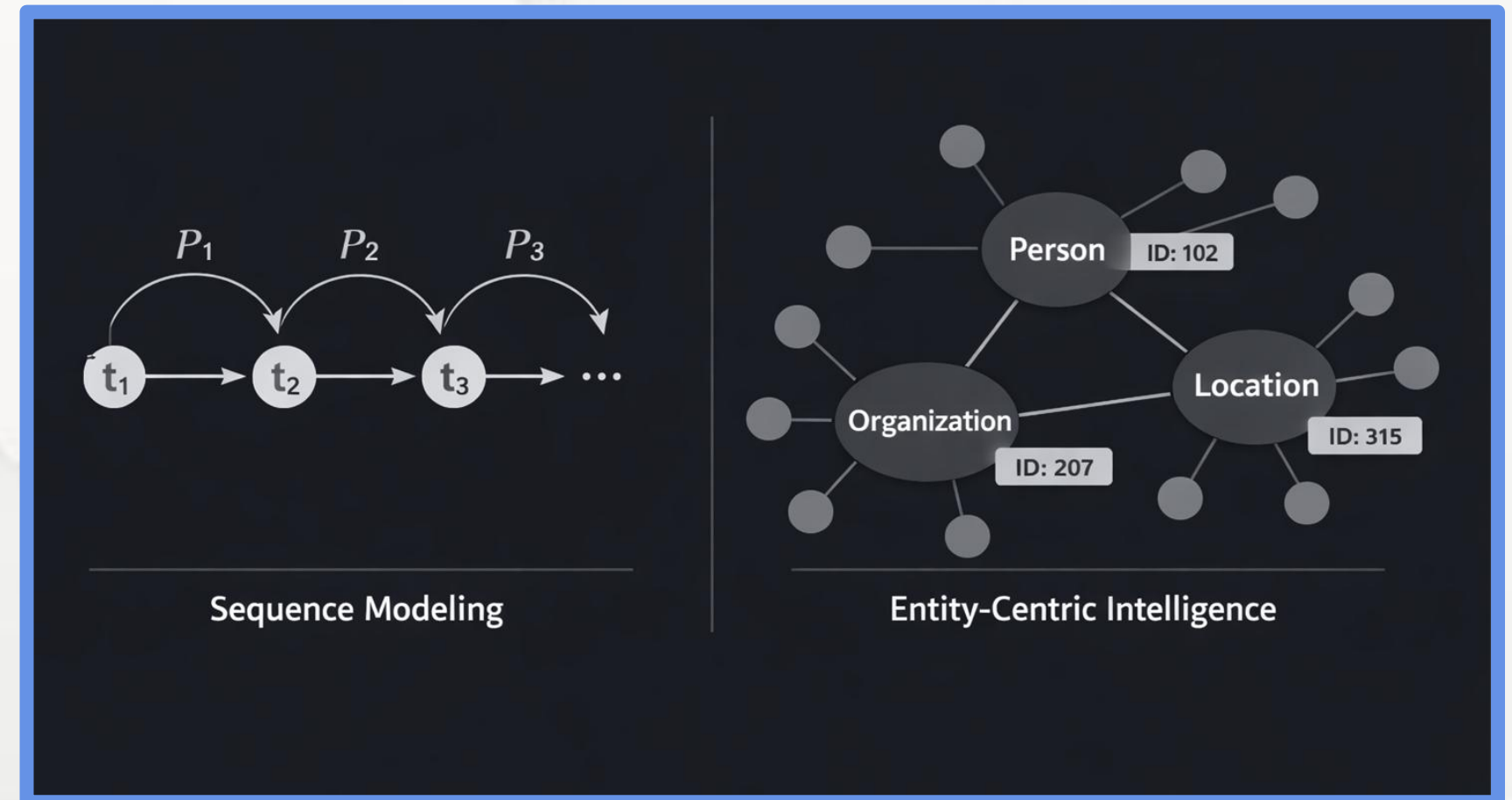
- 4.2 Named Entity Recognition (NER): Identification and Classification
- 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity
- 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases
- 4.5 Advanced Domain-Specific Entity Extraction
- 4.6 Summary and The Entity Lifecycle



Chapter 4.1 Introduction to Entity-Centric NLP

From Sequence Modeling to Entity-Centric Intelligence

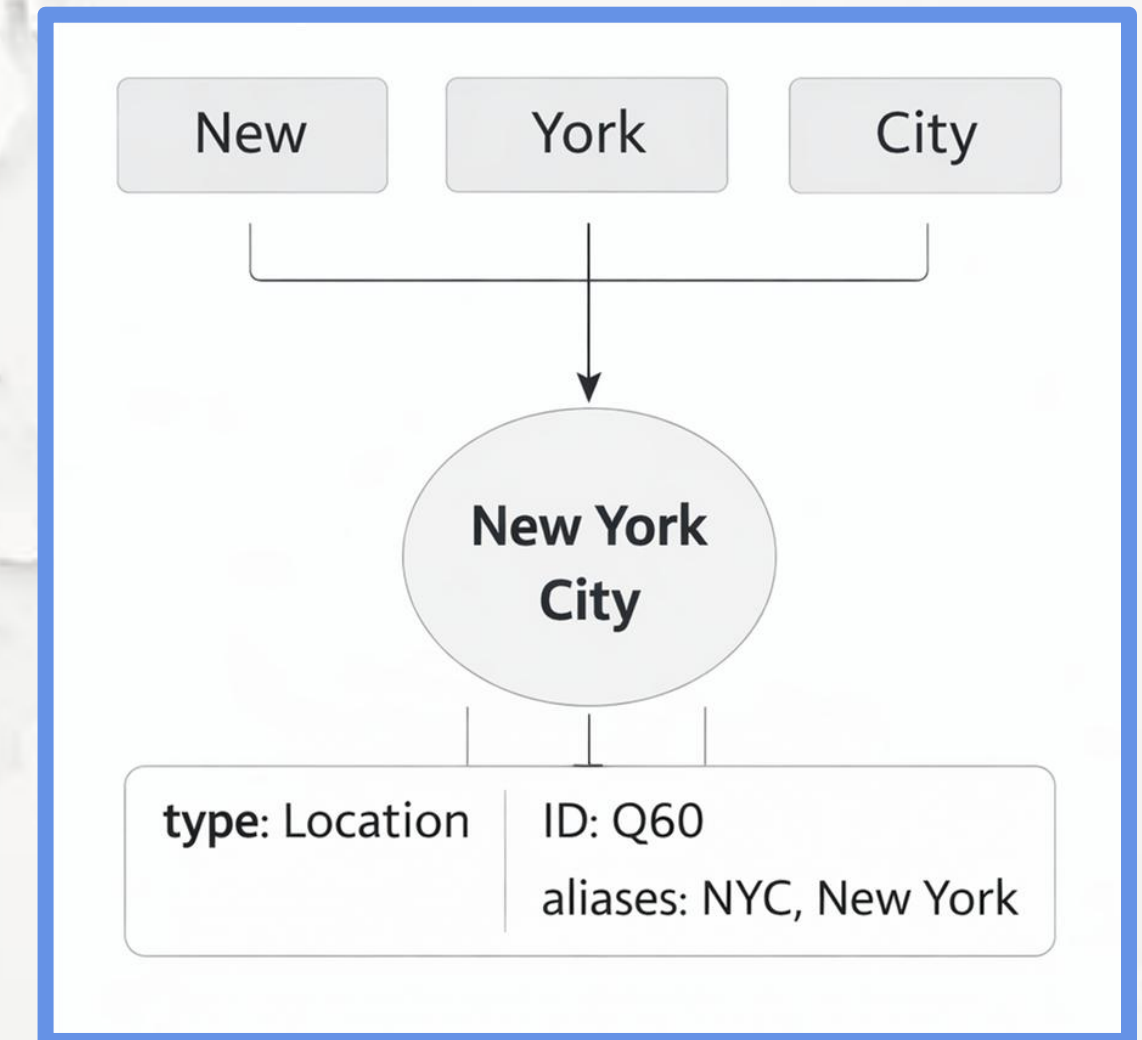
- Traditional NLP models language as token sequences and probabilities.
- High-reliability AI requires stable, identifiable real-world objects.
- Entity-Centric NLP shifts from linguistic tokens to knowledge objects.
- Entities enable tracking, memory, and reasoning across documents.
- This marks the transition from fluent language to grounded intelligence.



Chapter 4.1 Introduction to Entity-Centric NLP

What Is an Entity? The Atom of Knowledge

- An entity is a unique, persistent real-world or domain-specific object.
- Unlike tokens, entities maintain identity across contexts and documents.
- Examples include people, locations, organizations, laws, or chemicals.
- Entities are the stable units stored in knowledge graphs and databases.
- They enable long-term memory and cross-document reasoning.



4.1 Introduction to Entity-Centric NLP

4.2 Named Entity Recognition (NER): Identification and Classification

4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

4.5 Advanced Domain-Specific Entity Extraction

4.6 Summary and The Entity Lifecycle



Chapter 4.2 NER: Identification and Classification (Index)

What Is Named Entity Recognition?

- NER is a sequence tagging task over tokenized text.
- Each token is assigned a label indicating entity membership.
- Labels encode both boundaries and semantic categories.
- The output transforms raw text into structured entity spans.
- NER provides the foundation for all downstream entity tasks.



Chapter 4.2 NER: Identification and Classification (Index)

Standard Entity Types and Taxonomies

- NER systems classify entities into predefined semantic categories.
- Core universal types form the base of most industrial systems.
- Extended taxonomies adapt NER to specific domains.
- Correct typing is critical for downstream reasoning and linking.

Extended / domain types

- PER – People and individuals.
- ORG – Organizations and institutions.
- LOC – Locations (non-political).
- GPE – Countries, cities, geopolitical regions.

Core entity categories

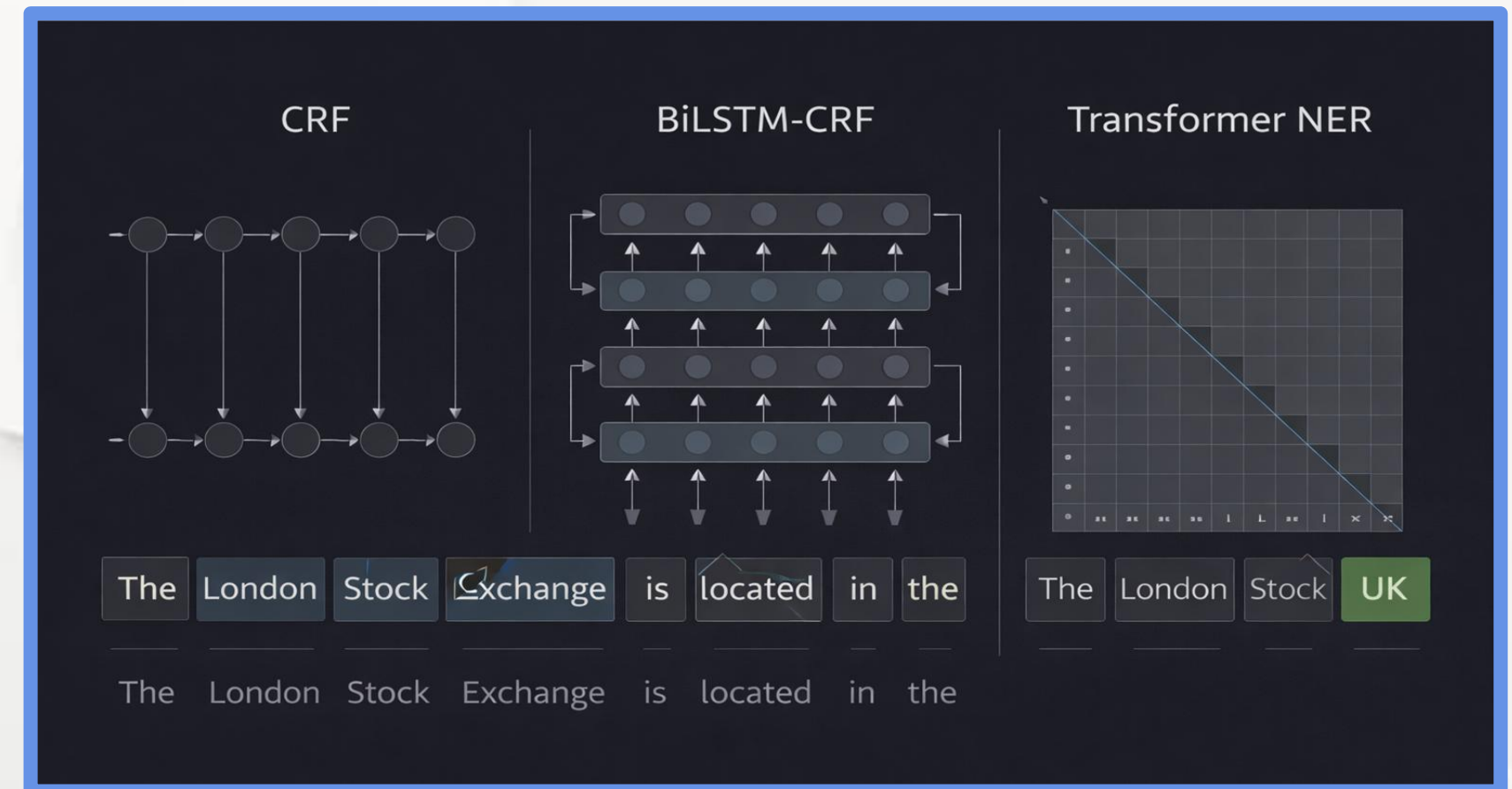
- PER – People and individuals.
- ORG – Organizations and institutions.
- LOC – Locations (non-political).
- GPE – Countries, cities, geopolitical regions.



Chapter 4.2 NER: Identification and Classification (Index)

Sequence Tagging Approaches

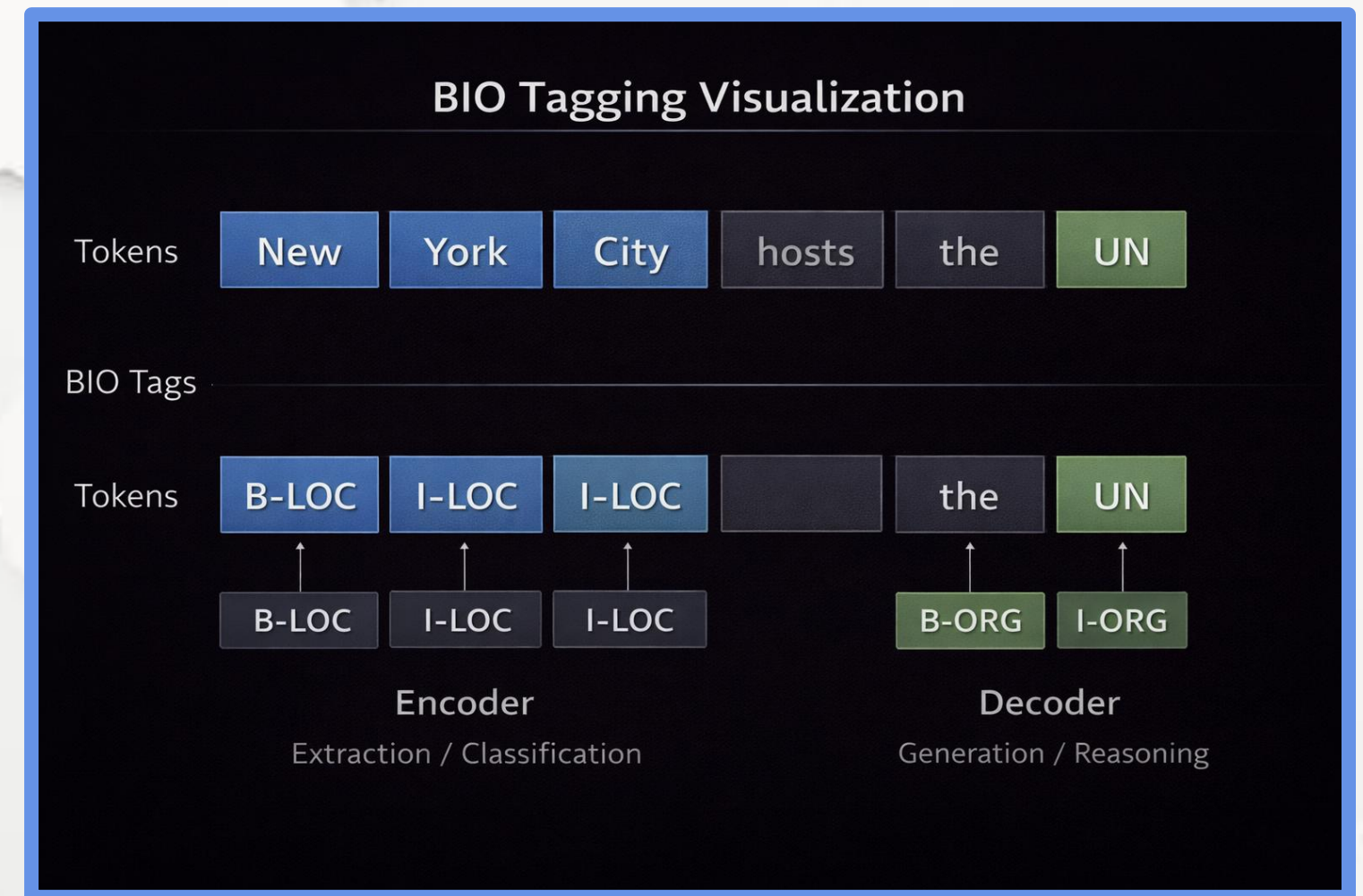
- NER is framed as a sequence labeling problem.
- Models predict a label for each token in context.
- Approaches evolved from probabilistic to deep neural architectures



Chapter 4.2 NER: Identification and Classification (Index)

Annotation Schemes: BIO / IOB Tagging

- Multi-token entities require boundary-aware labeling.
- BIO (or IOB) is the most widely used annotation scheme.
- Each token receives a boundary + type label.



Chapter 4.2 NER: Identification and Classification (Index)

Evaluation Metrics for NER

- NER evaluation is span-based, not token-based.
- Correct prediction requires exact boundary + label match.
- Partial matches count as errors.

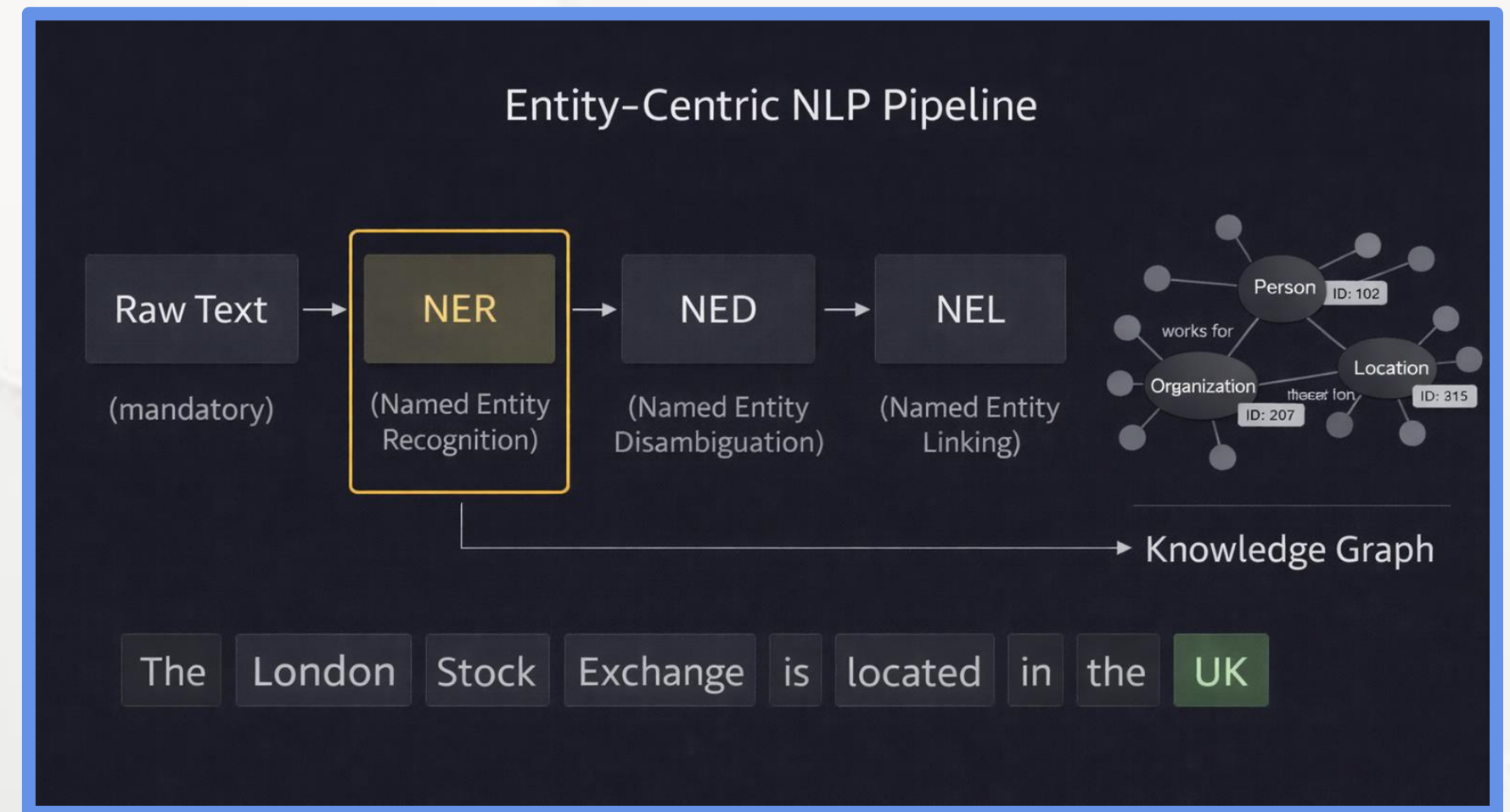
NER Evaluation	
	Gold entity spans Predicted entity spans
True Positive	London Stock Exchange = Located in
False Positive	London Stock Market ≠
False Negative	UK ≠ US
The London Stock Exchange is located in the UK	



Chapter 4.2 NER: Identification and Classification (Index)

Why NER Is the Foundation of Entity-Centric NLP

- NER is the entry point to entity intelligence.
- All downstream tasks depend on accurate detection.
- Errors propagate if NER fails.
- NER → NED → NEL → Relation Extraction → Knowledge Graphs



4.1 Introduction to Entity-Centric NLP

4.2 Named Entity Recognition (NER): Identification and Classification

4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

4.5 Advanced Domain-Specific Entity Extraction

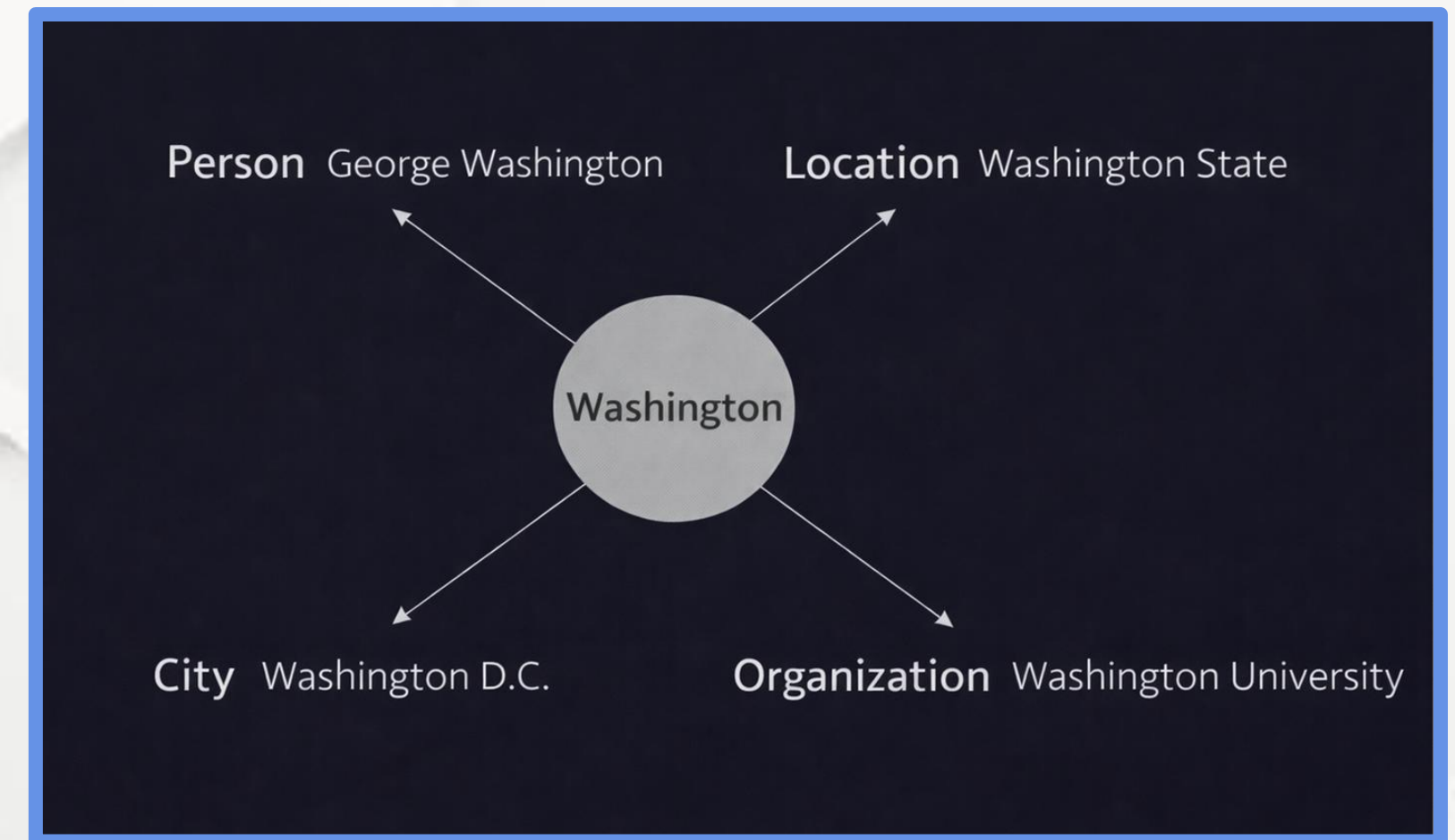
4.6 Summary and The Entity Lifecycle



Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

The Problem of Entity Ambiguity

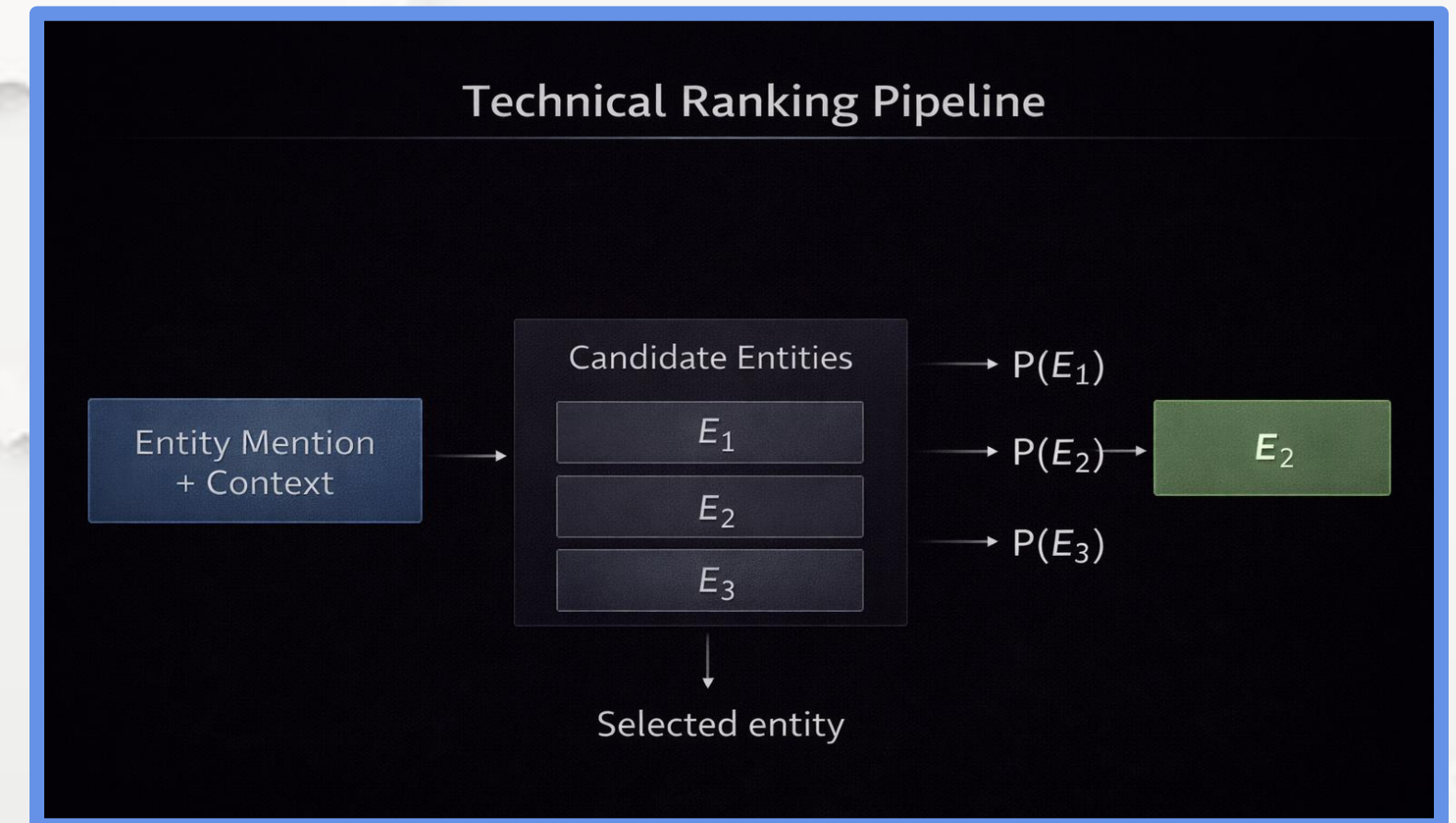
- The same surface name can refer to multiple real entities.
- Ambiguity is unavoidable in natural language.
- NER alone cannot determine the correct referent.
- Unresolved ambiguity leads to factual errors.



Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

What Is Named Entity Disambiguation (NED)

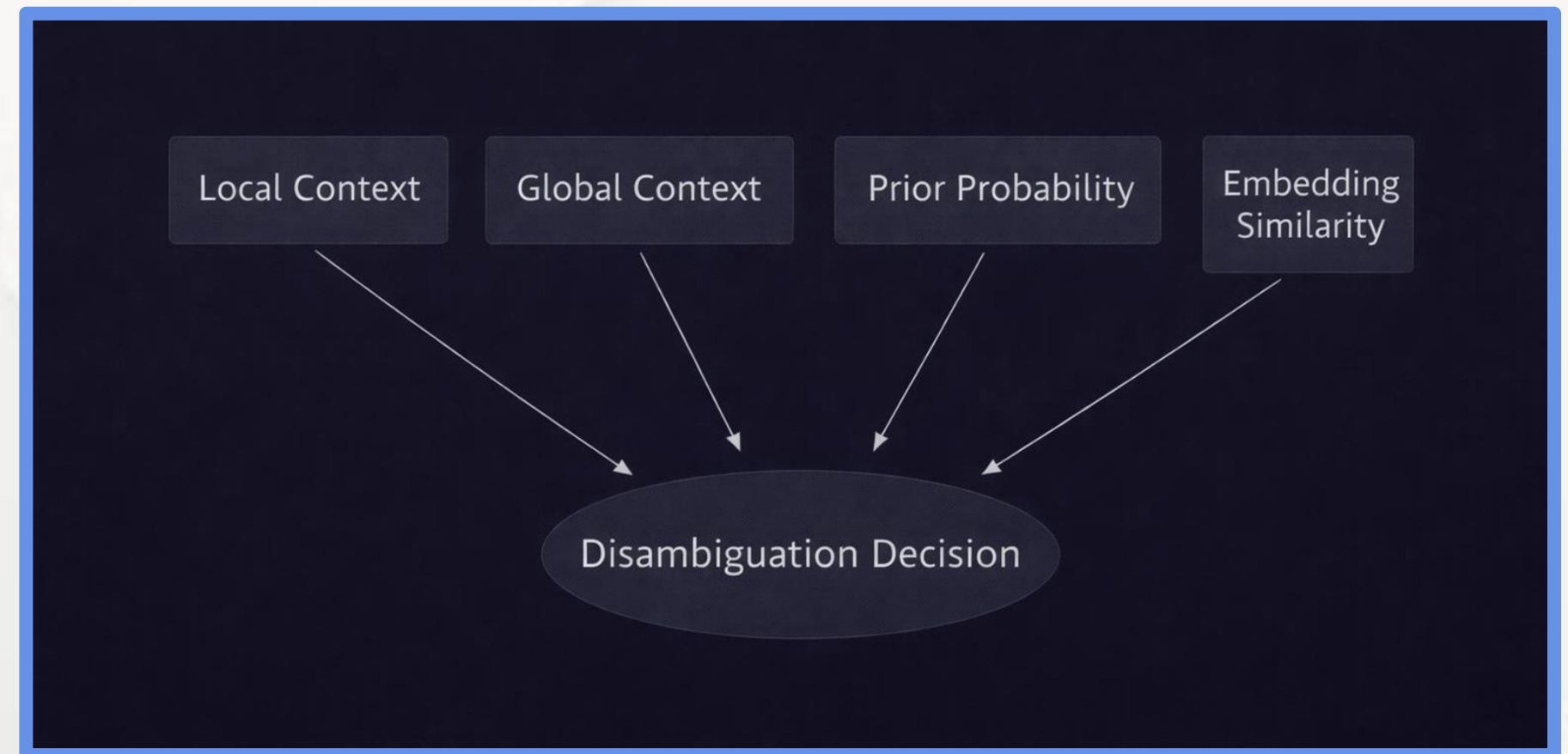
- NED selects the correct entity among candidates.
- It is modeled as a ranking problem.
- Input: mention + context.
- Output: single most probable real-world entity.



Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

Signals Used in Entity Disambiguation

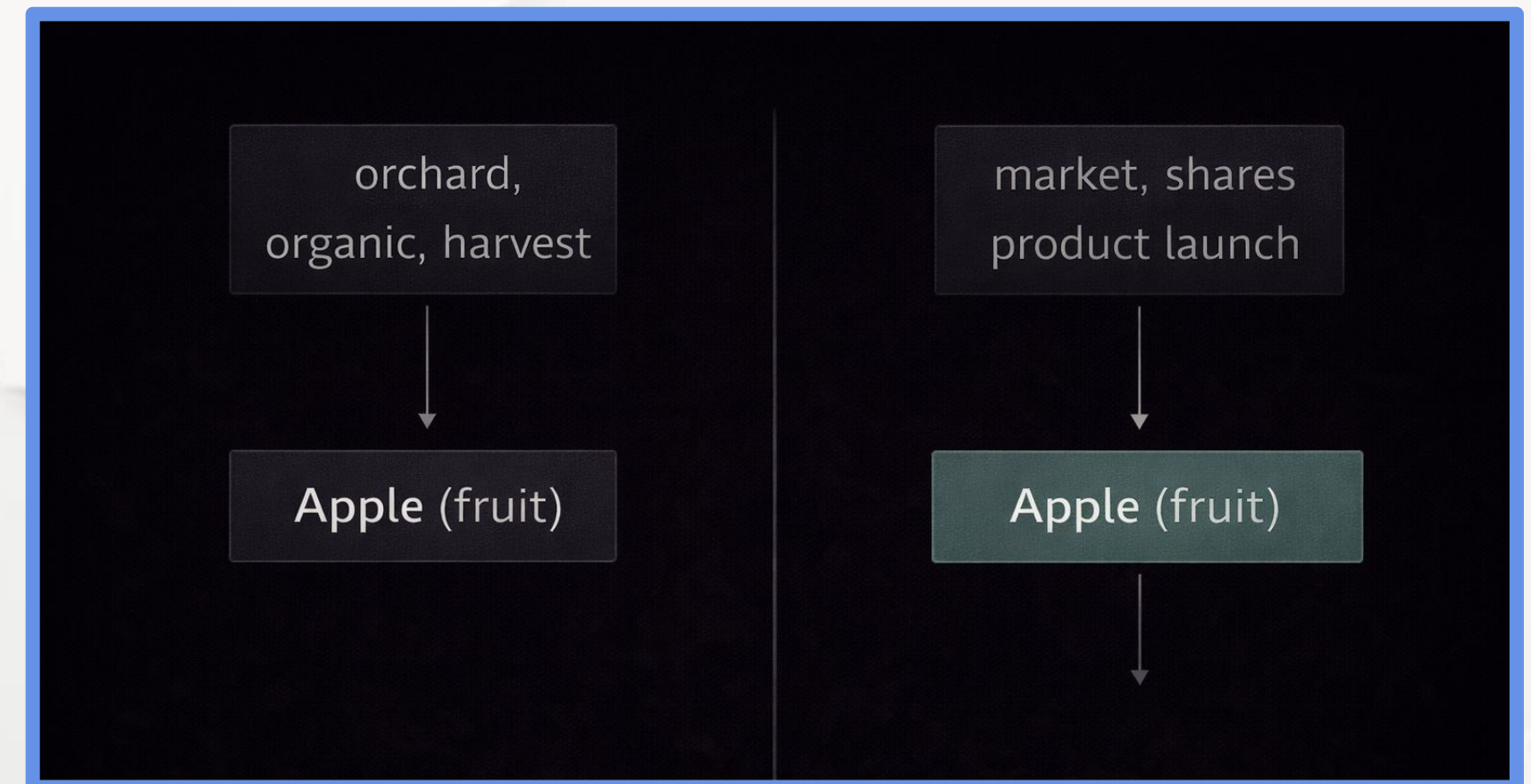
- Local context around the mention.
- Global document coherence.
- Prior probability from knowledge bases.
- Semantic similarity in embedding space.



Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

Example: Resolving Ambiguous Mentions

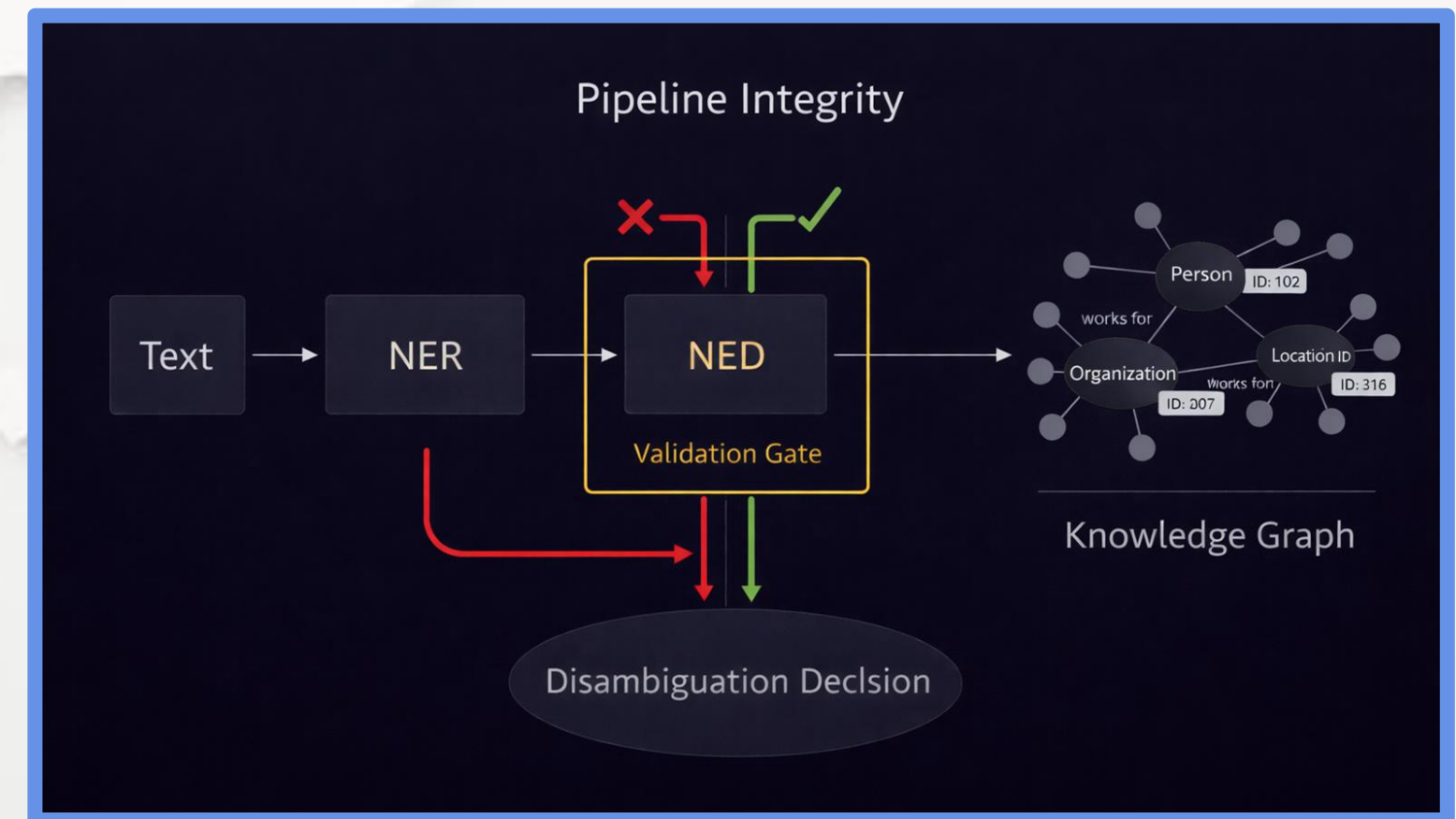
- Context determines the entity meaning.
- Semantic cues guide the resolution.
- Same word, different real-world entities.
- Correct resolution enables safe reasoning.



Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

Why NED Is Critical for Knowledge Systems

- Prevents incorrect facts entering knowledge graphs.
- Enables precise analytics and reasoning.
- Required before entity linking (NEL).
- Core reliability layer in enterprise AI.



4.1 Introduction to Entity-Centric NLP

4.2 Named Entity Recognition (NER): Identification and Classification

4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

4.5 Advanced Domain-Specific Entity Extraction

4.6 Summary and The Entity Lifecycle



Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

What Is Named Entity Linking (NEL)

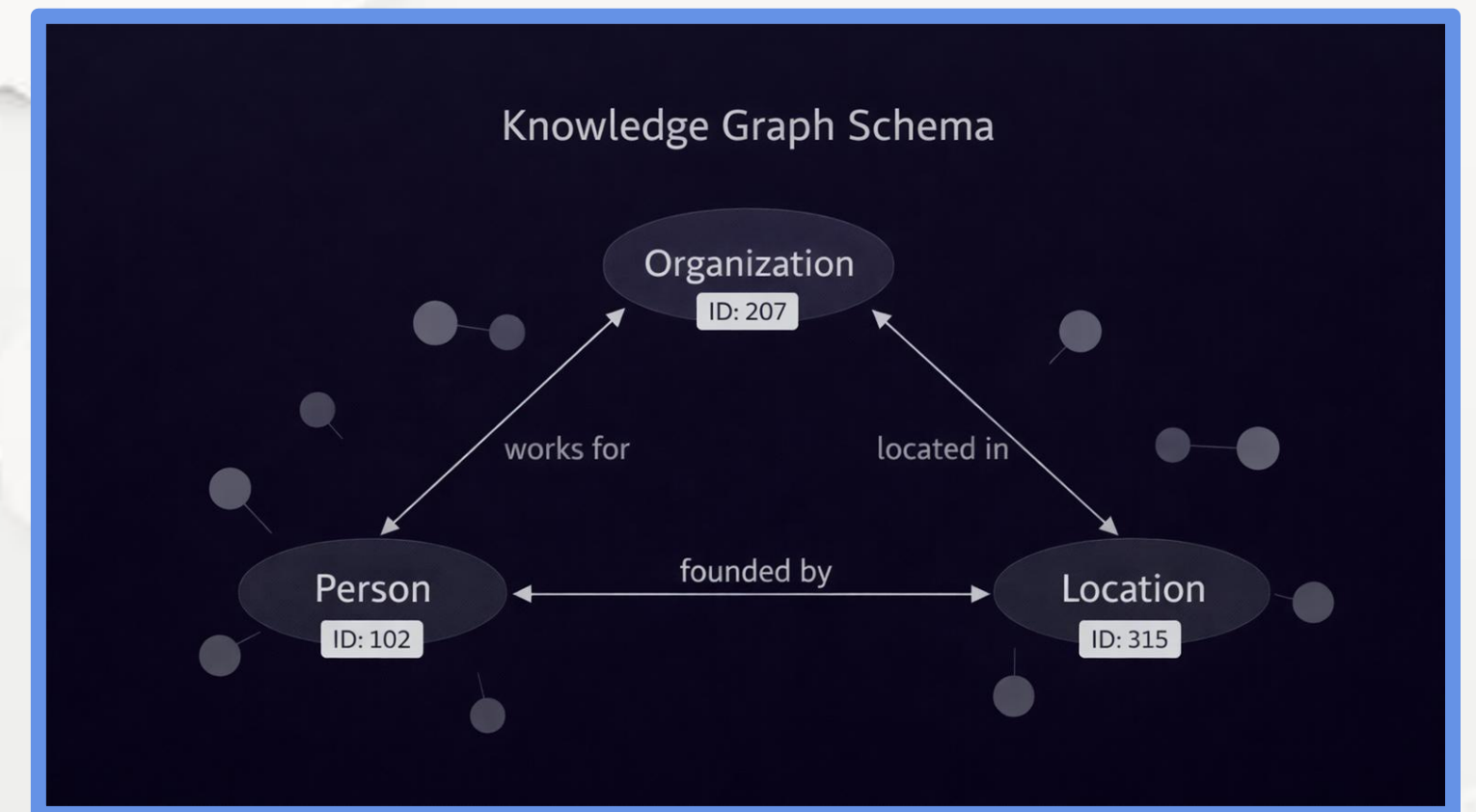
- NEL connects text mentions to real-world entities.
- Each entity is resolved to a unique identifier.
- Eliminates ambiguity across aliases and synonyms.
- Enables persistent, machine-readable references.



Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

Knowledge Bases and Unique Identifiers

- Knowledge Bases store structured world facts.
- Entities are identified by global IDs.
- Relations are stored as subject–predicate–object triples.
- Enables reasoning beyond raw text.

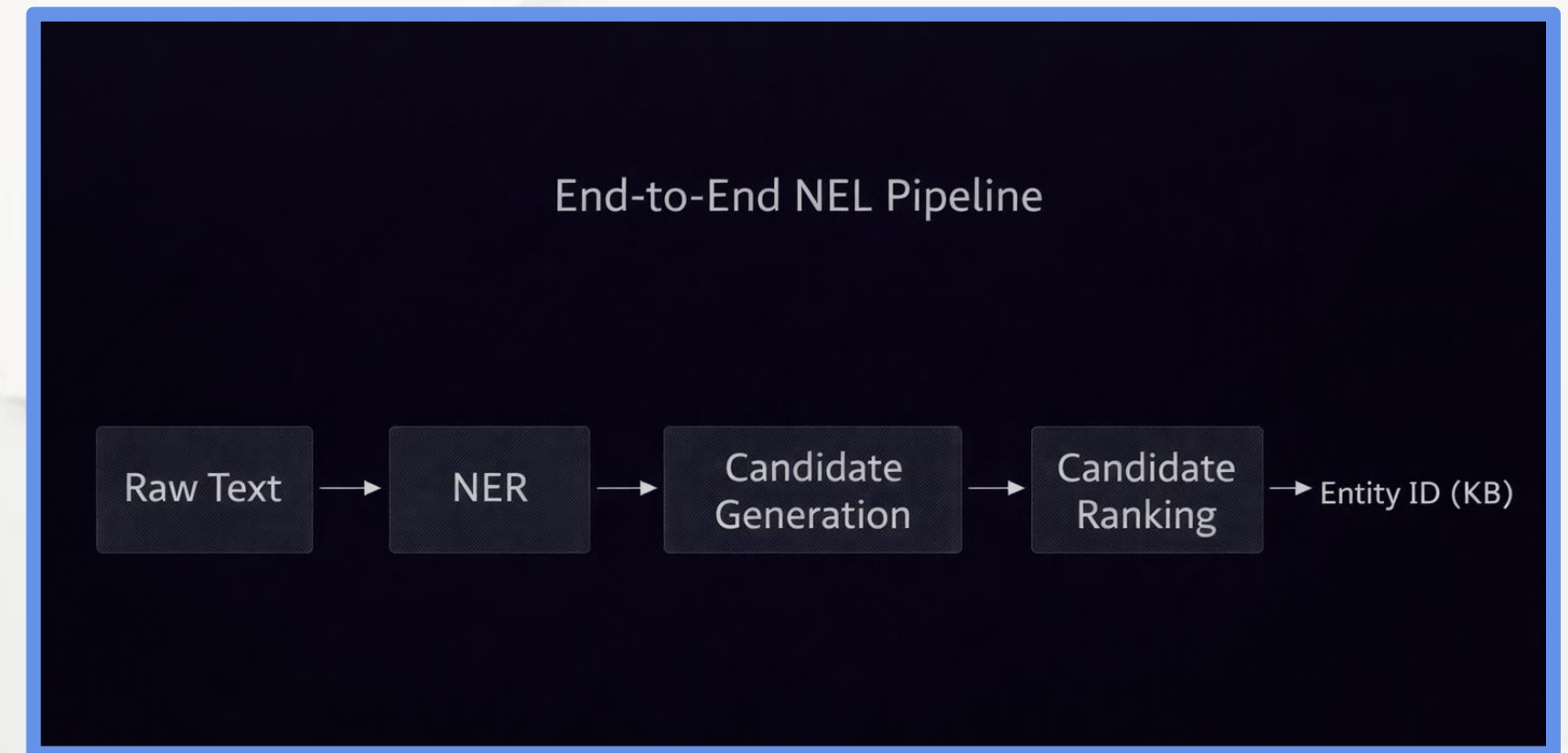


M04

Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

The NEL Engineering Pipeline

- Detect entity mentions (NER).
- Generate candidate entities.
- Rank candidates using context (NED).
- Link to knowledge base ID.
- Handle unknown entities (NIL).



CONFianza23
Inteligencia y Seguridad

Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

NED vs NEL: Key Differences

- NED selects the correct candidate.
- NEL implements the full linking system.
- NED is a decision task.
- NEL is an end-to-end architecture.

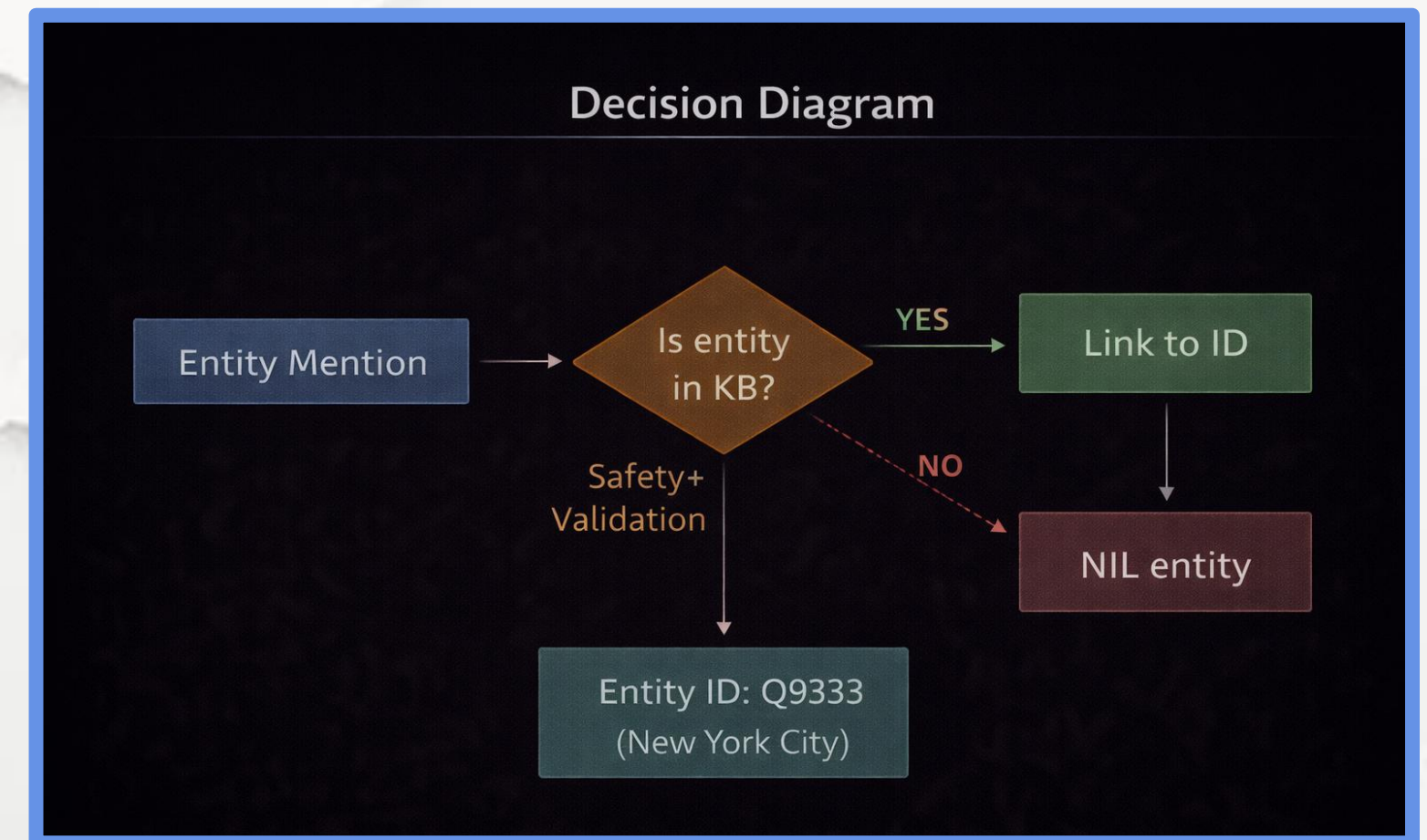
NED vs. NEL: Key Differences	
NED (Named Entity Disambiguation)	NEL (Named Entity Linking)
Scope Focuses on identifying which known entity an ambiguous mention refers to WITHIN a document/context	- Connects mentions of entities in entries, gl. Wikipedia, Widarat) - Connects mentions of entities in text text to entries in knowledge base (e.gl. Wikipedia, Widarat)
Output A specific entity ID or reference from preeffined set	A URI/URL or unique identifier to a knowledge base entry
Complexity Context-dependent; requires local analysis	Higher Requires robust knowledge base lookup & entity resolution
Role in pipeline Often a sub-component of a larger NLP system	Often a downtoan task from NED; enriches text with external knowledge



Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

NIL Detection and Real-World Constraints

- Not all entities exist in knowledge bases.
- Systems must detect NIL cases safely.
- Prevents hallucinated or false entities.
- Critical for private or emerging entities.

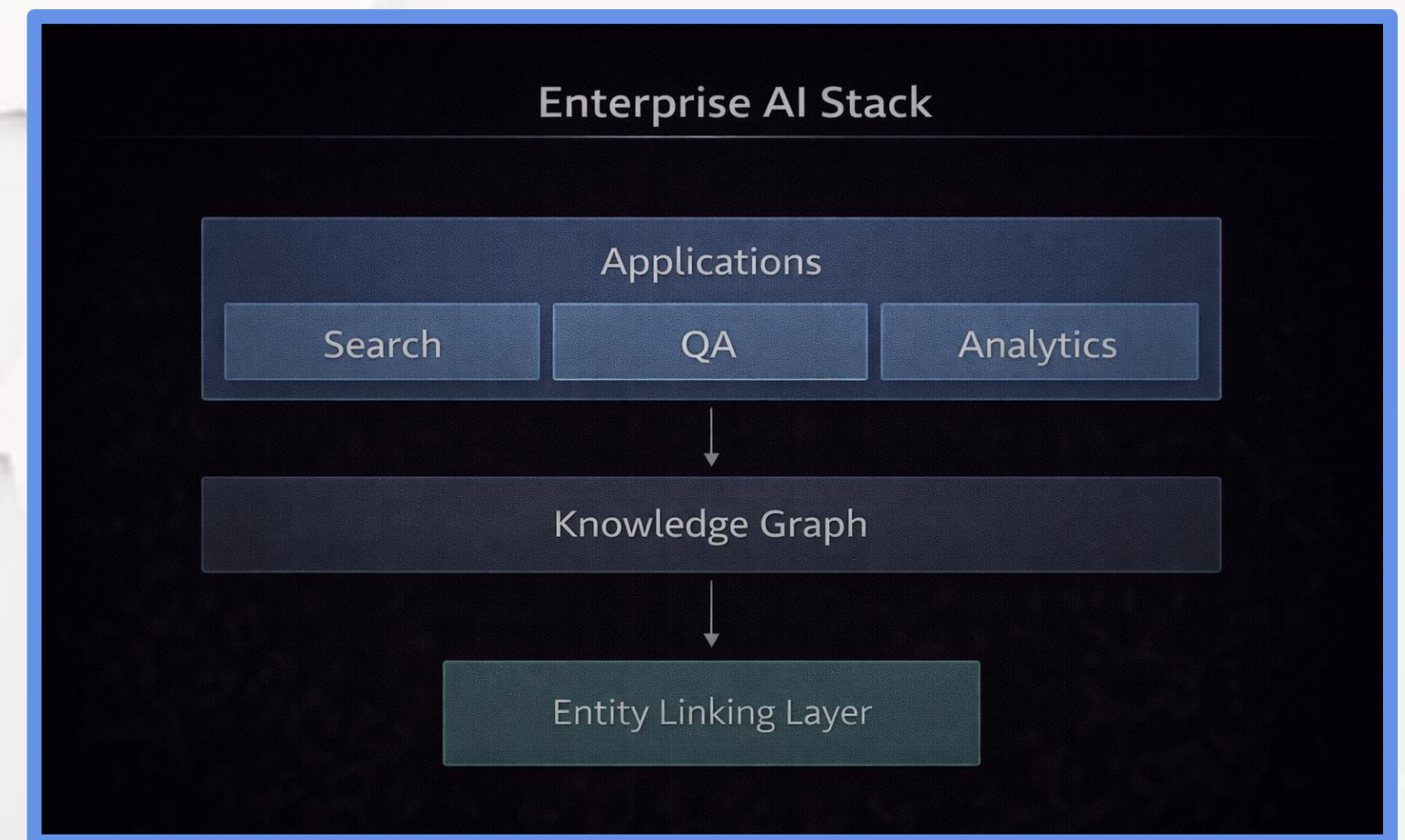


M04

Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

Why NEL Is Essential for Reliable AI

- Grounds language in factual reality.
- Enables explainable reasoning.
- Supports analytics and traceability.
- Foundation for enterprise knowledge graphs.



CONFIANZA23
Inteligencia y Seguridad

4.1 Introduction to Entity-Centric NLP

4.2 Named Entity Recognition (NER): Identification and Classification

4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

4.5 Advanced Domain-Specific Entity Extraction

4.6 Summary and The Entity Lifecycle



Chapter 4.5 Advanced Domain-Specific Entity Extraction

Why Domain-Specific Entities Matter

- General NER fails in specialized domains.
- Domain entities are longer and more complex.
- Labels are highly specific and non-standard.
- Precision is critical in regulated environments.

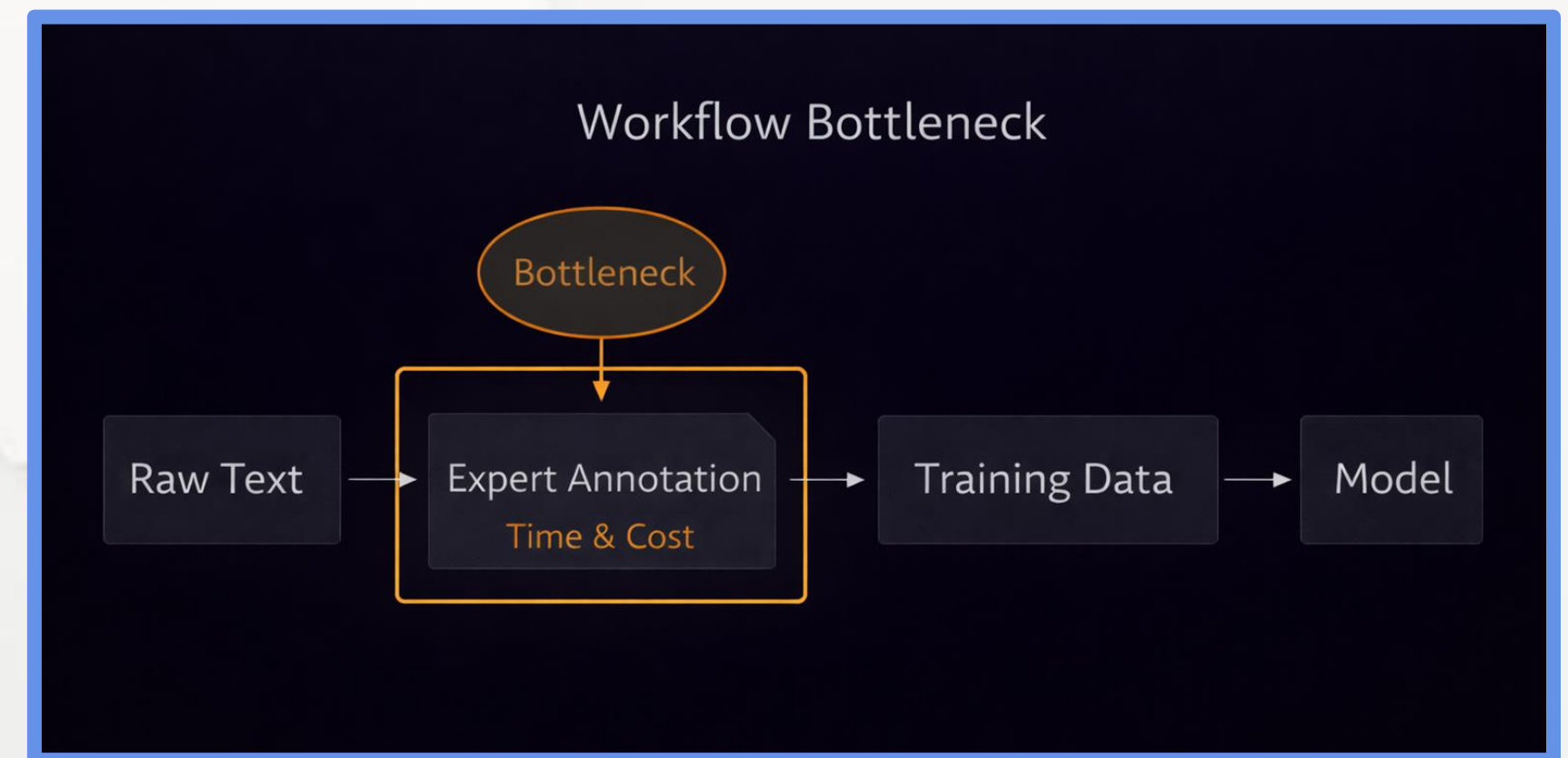
General NER	Domain NER
• Person • Location • Organization	• Legal Clause • Chemical Compound • Medical Condition
• Person • Location • Organization	• Legal Clause • Chemical Compound • Medical Condition



Chapter 4.5 Advanced Domain-Specific Entity Extraction

The Annotation Bottleneck

- Domain annotation requires expert knowledge.
- Manual labeling is slow and expensive.
- Data scarcity limits supervised training.
- Annotation quality impacts downstream accuracy.

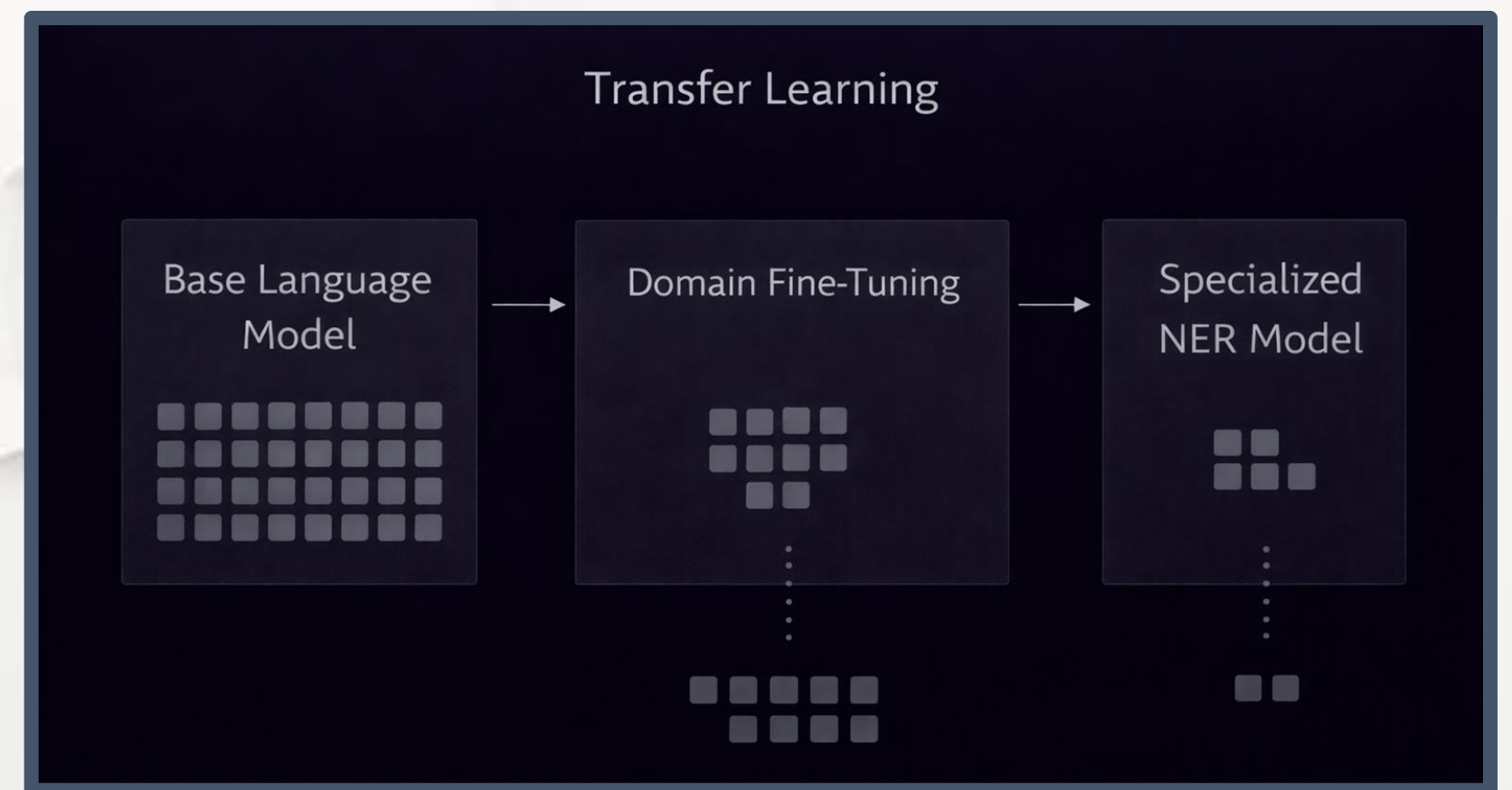


M04

Chapter 4.5 Advanced Domain-Specific Entity Extraction

Transfer Learning for Specialized NER

- Domain annotation requires expert knowledge.
- Manual labeling is slow and expensive.
- Data scarcity limits supervised training.
- Annotation quality impacts downstream accuracy.



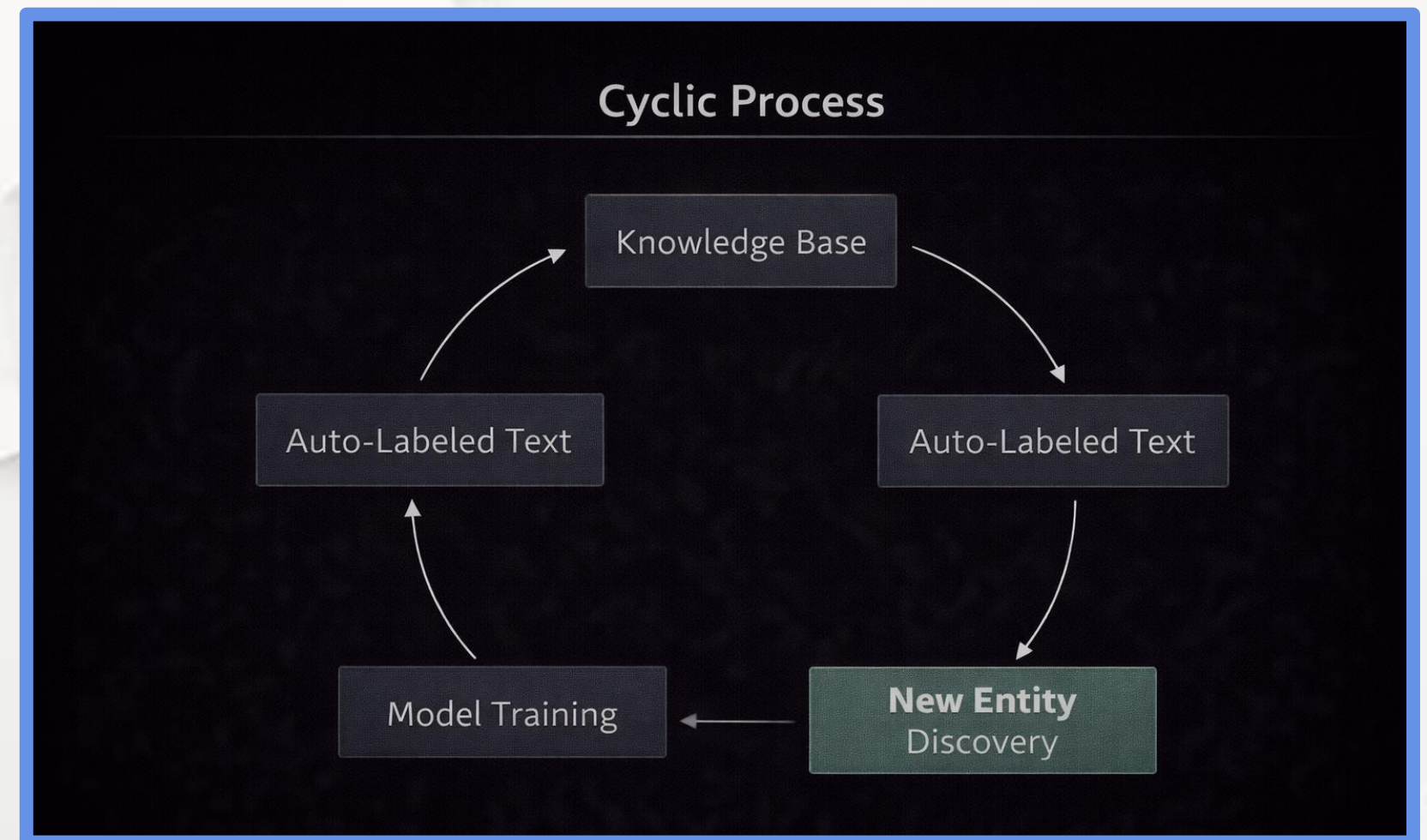
CONFIANZA23
Inteligencia y Seguridad

M04

Chapter 4.5 Advanced Domain-Specific Entity Extraction

Distant Supervision and Bootstrapping

- Use existing databases as weak labels.
- Automatically annotate large corpora.
- Iteratively expand entity coverage.
- Produces scalable “silver” datasets.



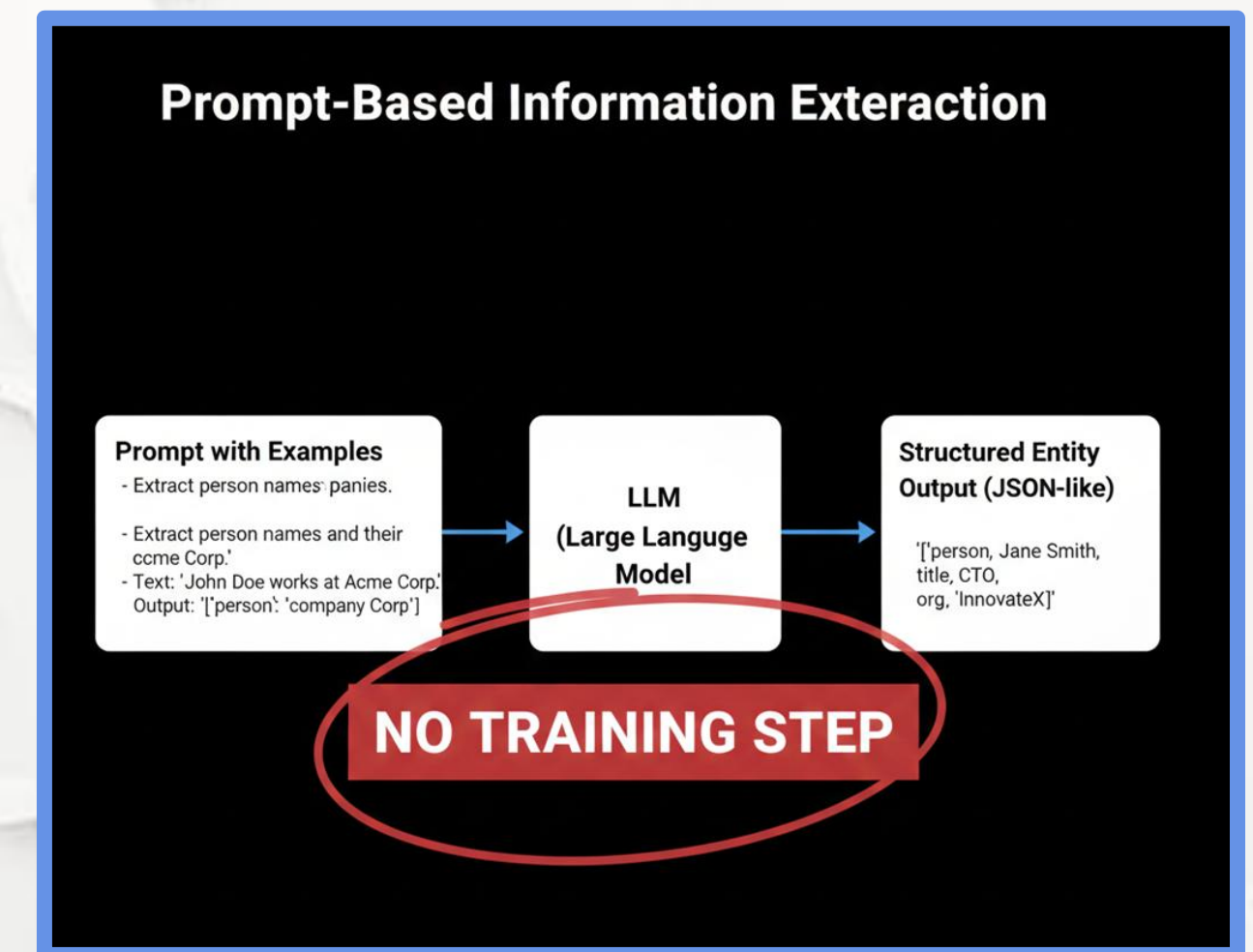
CONFIANZA23
Inteligencia y Seguridad

M04

Chapter 4.5 Advanced Domain-Specific Entity Extraction

LLM Prompting for Entity Extraction

- Zero-shot extraction via instructions.
- Few-shot examples improve precision.
- Rapid adaptation without retraining
- Outputs used as silver labels.

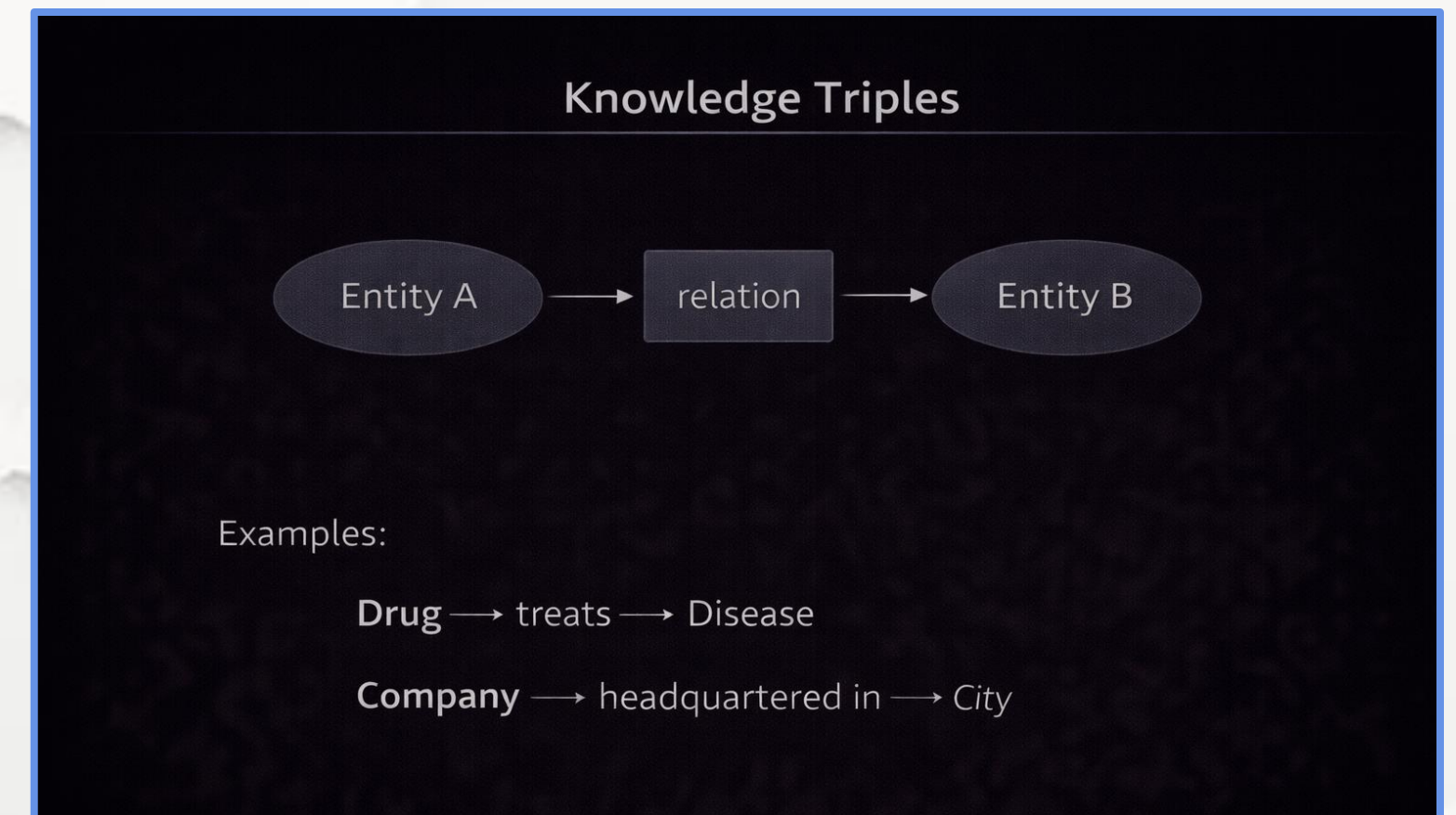


CONFIANZA23
Inteligencia y Seguridad

Chapter 4.5 Advanced Domain-Specific Entity Extraction

Relation Extraction as the Next Step

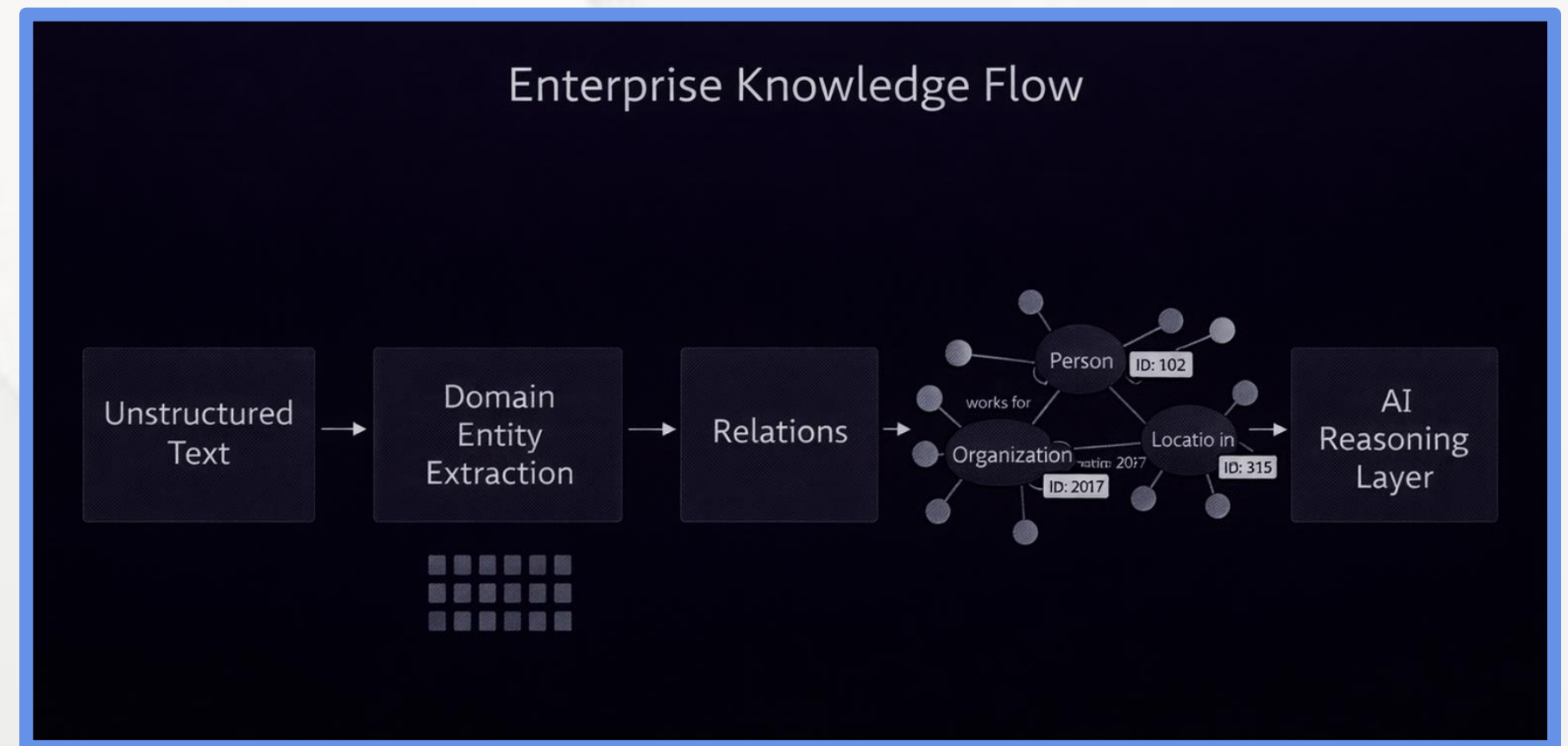
- Entities alone are not sufficient.
- Relations capture semantic structure.
- Output expressed as triples.
- Enables dynamic knowledge graphs.



Chapter 4.5 Advanced Domain-Specific Entity Extraction

Why Domain-Specific Extraction Enables Reasoning

- Transforms documents into structured data.
- Supports automated reasoning systems.
- Enables real-time knowledge updates.
- Foundation for decision intelligence.



4.1 Introduction to Entity-Centric NLP

4.2 Named Entity Recognition (NER): Identification and Classification

4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

4.5 Advanced Domain-Specific Entity Extraction

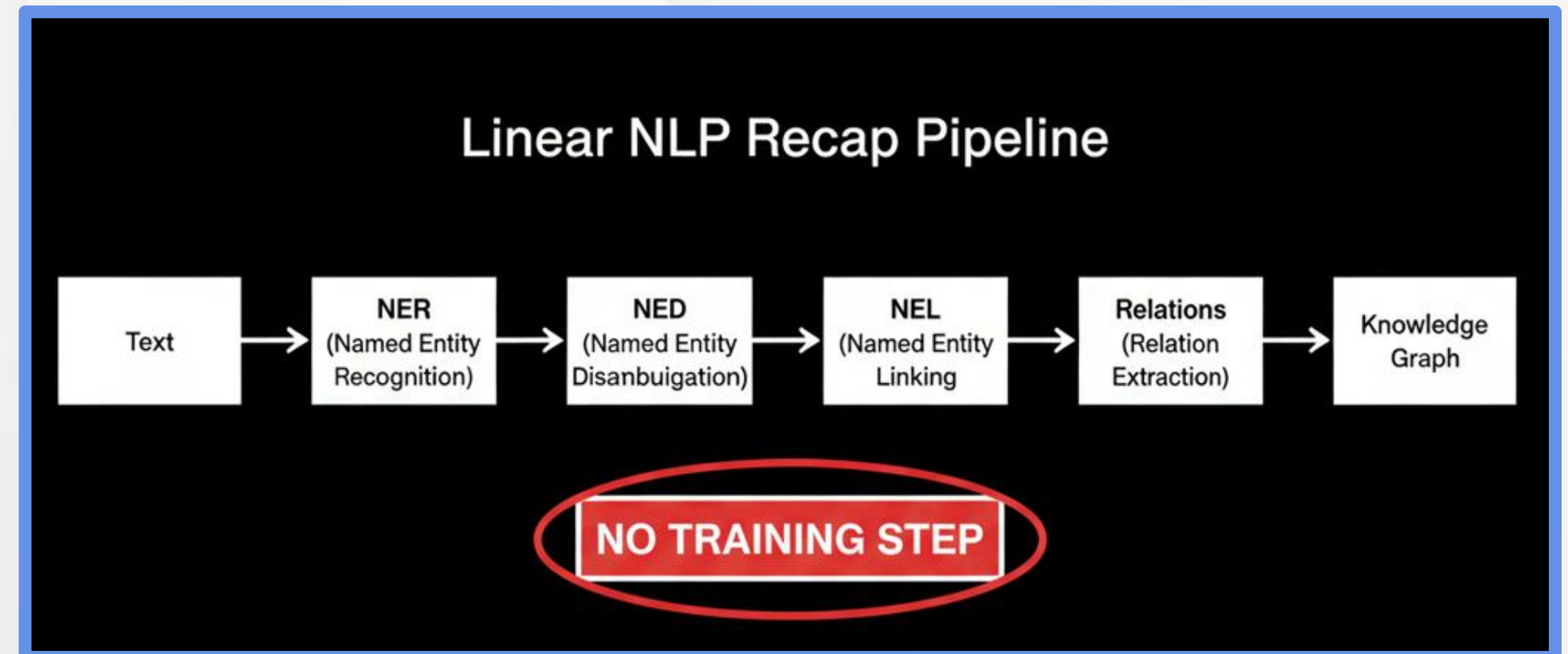
4.6 Summary and The Entity Lifecycle



Chapter 4.6 Summary and The Entity Lifecycle

Recap of the Entity-Centric Pipeline

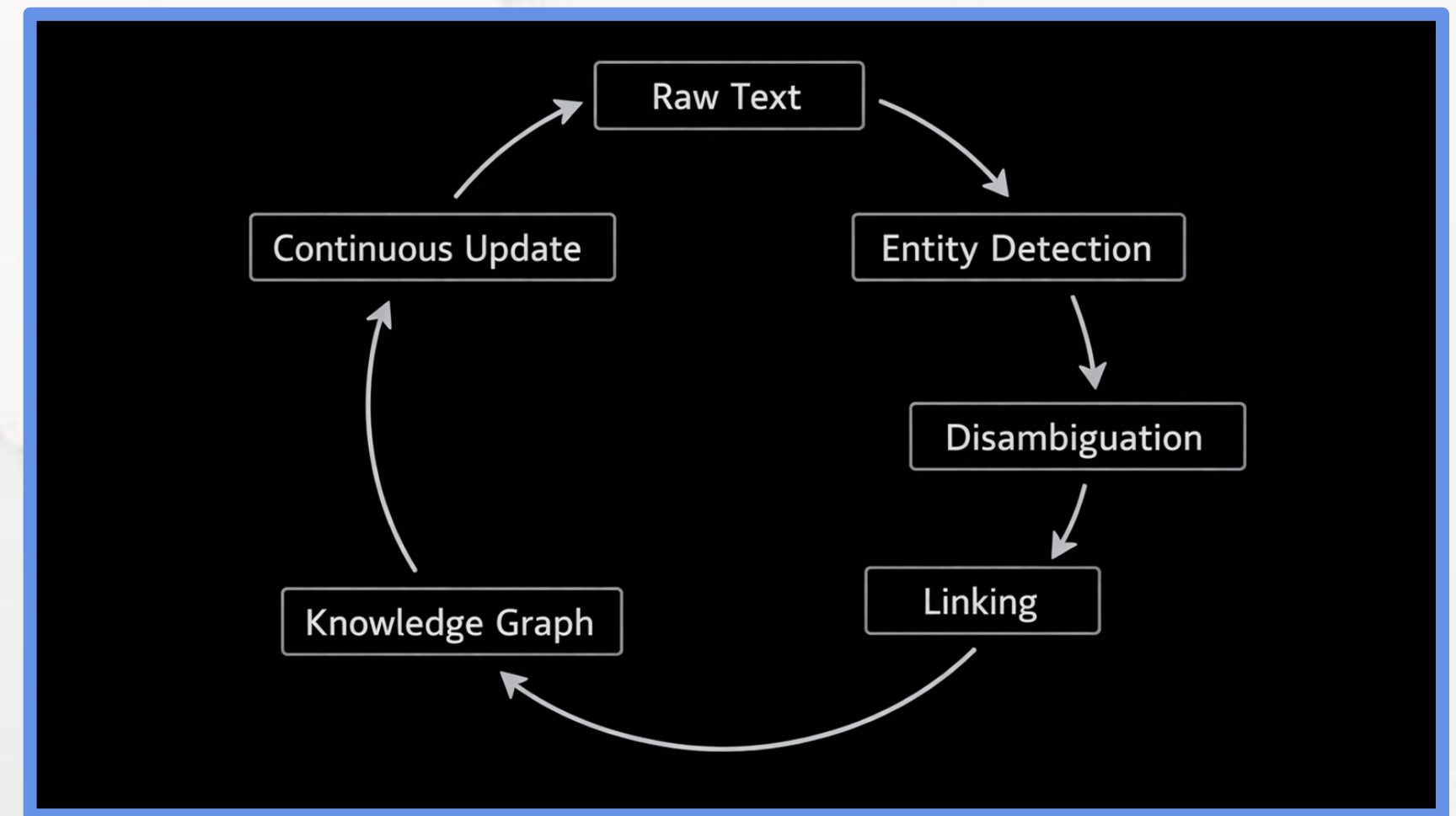
- Entities are the atomic units of meaning.
- NER identifies entity spans.
- NED resolves ambiguity.
- NEL grounds entities in knowledge bases.
- Relations connect entities semantically.



Chapter 4.6 Summary and The Entity Lifecycle

The Complete Entity Lifecycle

- Entities emerge from raw text.
- Entities are validated and linked.
- Entities evolve over time.
- Knowledge graphs are continuously updated.



Chapter 4.6 Summary and The Entity Lifecycle

Key Challenges in Real-World Entity Systems

- Low-resource and emerging entities.
- Ambiguous or overlapping mentions.
- Knowledge base incompleteness.
- Entity drift over time.

CHALLENGE MATRIX	
 Low-resource entities Limited information availability	 Ambiguity Contextual uncertainty
 KB gaps Missing knowledge in knowledge bases	 Temporal drift Changes over time

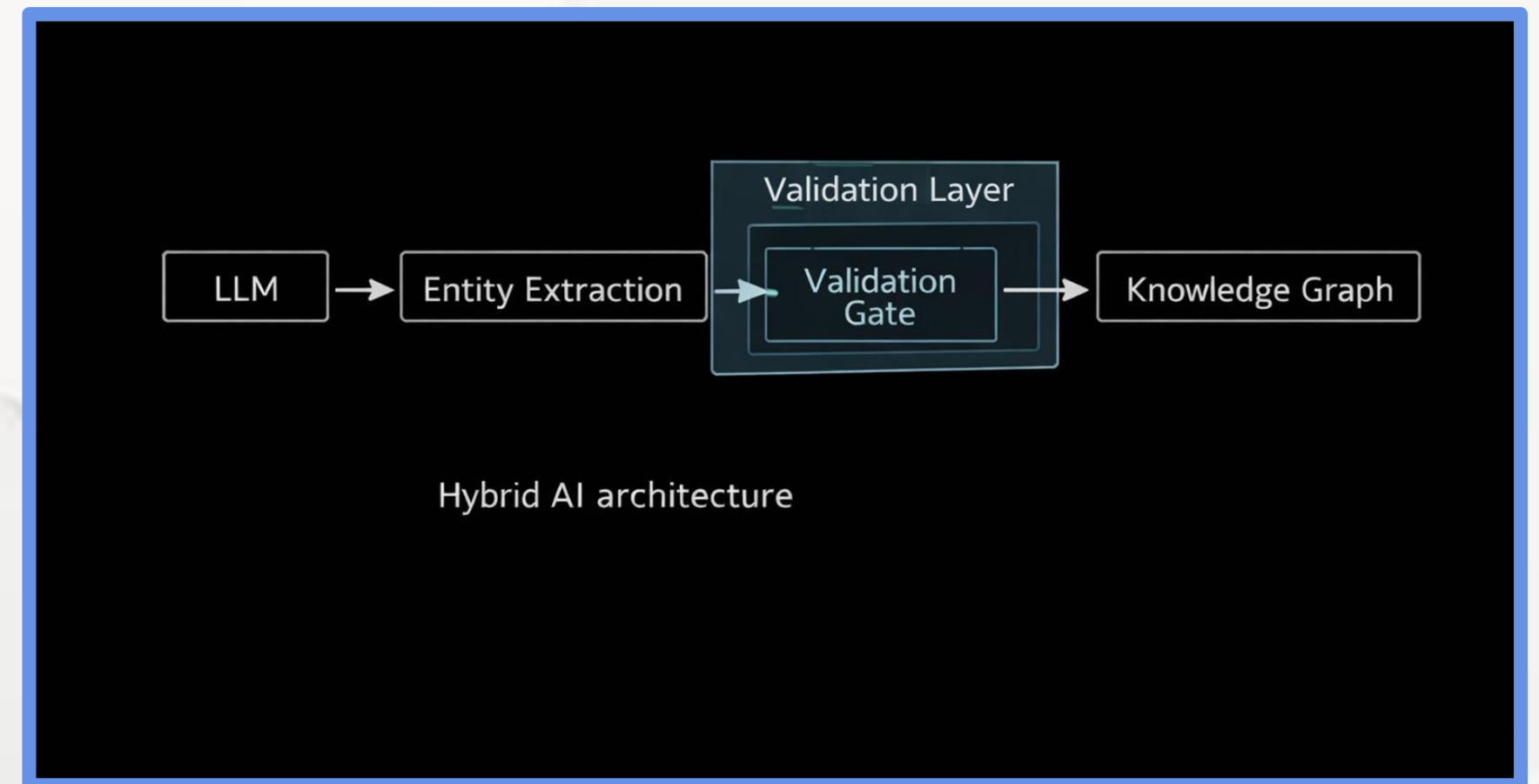


M04

Chapter 4.6 Summary and The Entity Lifecycle

Role of Generative Models in Entity NLP

- LLMs act as internal annotators.
- Enable zero-shot entity extraction.
- Assist relation discovery.
- Must be validated to prevent hallucinations.

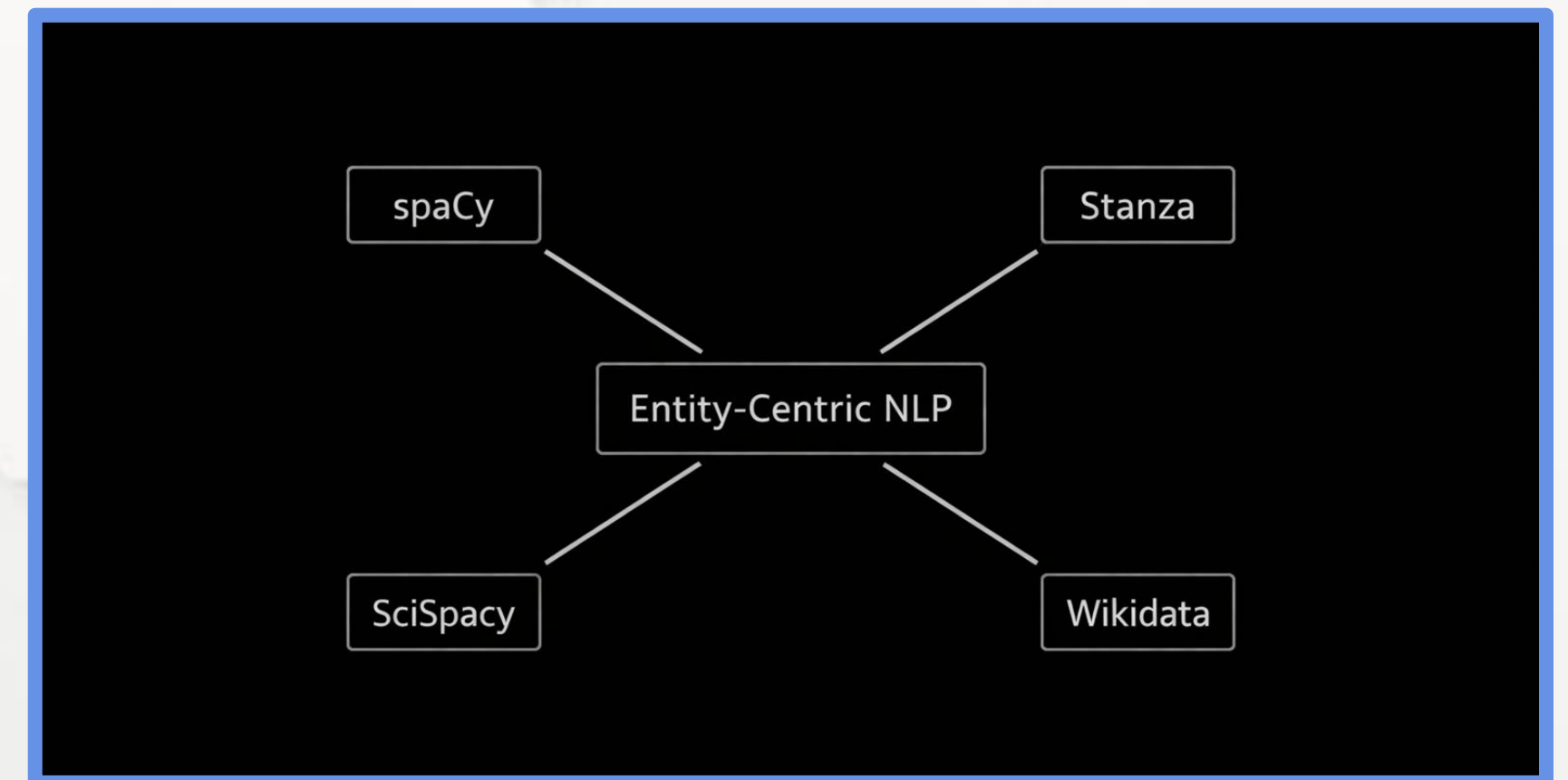


CONFianza23
Inteligencia y Seguridad

Chapter 4.6 Summary and The Entity Lifecycle

Practical Tools and Libraries

- spaCy: production-grade NER pipelines.
- Stanza: multilingual neural NLP.
- SciSpacy: biomedical entity extraction.
- Wikidata: global knowledge grounding.



M04

Chapter 4.6 Summary and The Entity Lifecycle

Final Takeaways

- Entities anchor language to reality.
- Entity grounding reduces hallucinations.
- Knowledge graphs enable explainability.
- Entity-centric NLP is essential for trusted AI.

CONCEPTUAL TAKEAWAY DIAGRAM

 Text probabilistic layer →



 Entity grounding layer →



 Reliable AI system

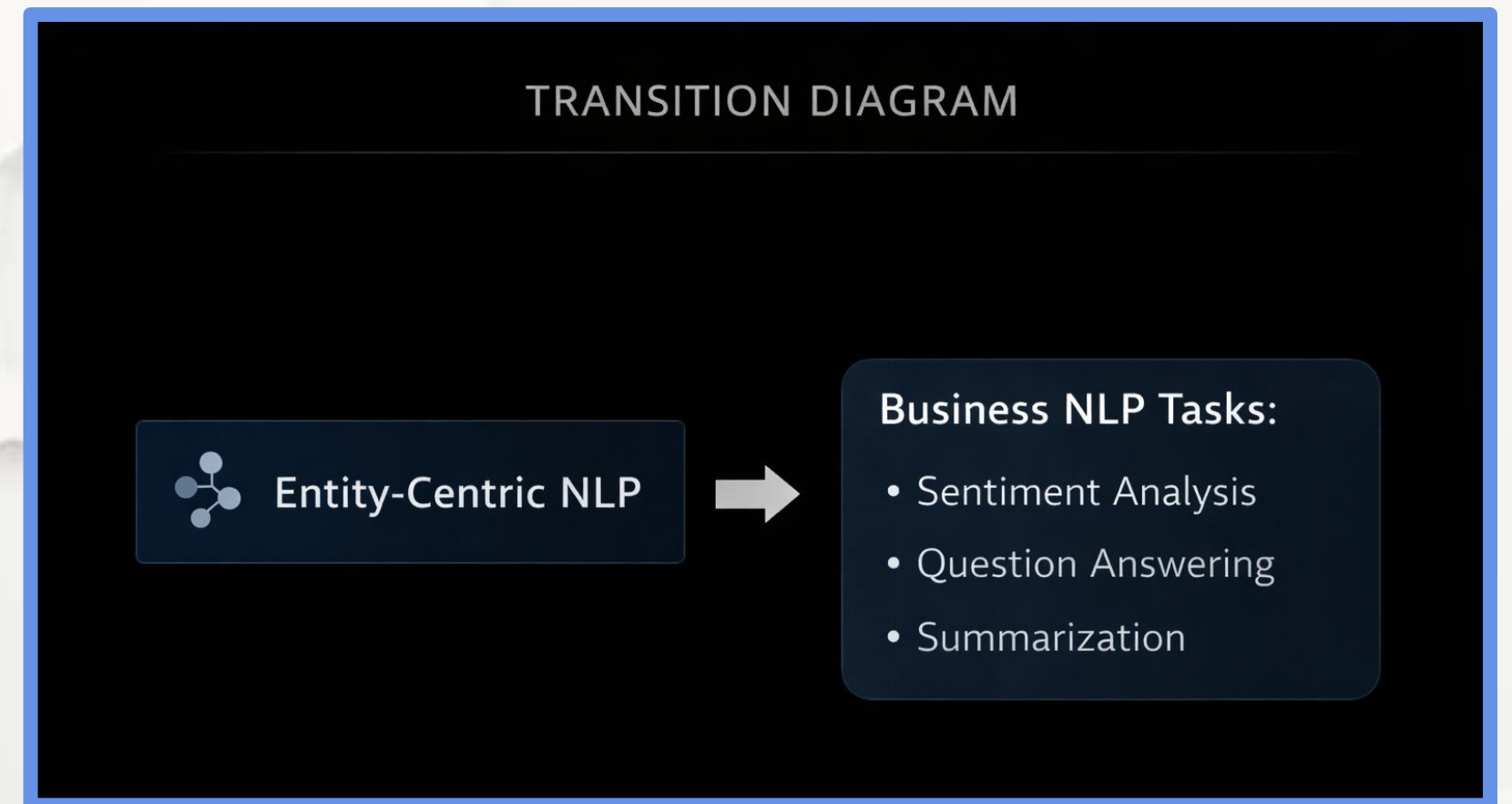


CONFIANZA23
Inteligencia y Seguridad

Chapter 4.6 Summary and The Entity Lifecycle

Transition to the Next Module

- Entity-centric NLP enables downstream tasks.
- Grounded entities improve analytics accuracy.
- Next: applying entities to business problems.
- Sentiment, QA, and summarization.



M04

Chapter 4.6 Summary and The Entity Lifecycle

M04 – Academic References and Research Resources

- Nadeau, D. & Sekine, S. (2007) A Survey of Named Entity Recognition and Classification
- Lafferty, J., McCallum, A., Pereira, F. (2001) Conditional Random Fields: Probabilistic Models for Segmenting and Labeling Sequence Data
- Collobert, R. et al. (2011) Natural Language Processing (Almost) from Scratch
- Bunescu, R. & Paşca, M. (2006) Using Encyclopedic Knowledge for Named Entity Disambiguation
- Ganea, O. & Hofmann, T. (2017) Deep Joint Entity Disambiguation with Local Neural Attention
- Shen, W. et al. (2015) Entity Linking with a Knowledge Base: Issues, Techniques, and Solutions



CONFIANZA23
Inteligencia y Seguridad



CONFIANZA23
Inteligencia y Seguridad